Final

Lecture-1

05.10.2026

Syllabus: i) Regular Expression } CT

ii) CFG

iii) Language Processing System

iv) Compiler Design } CT

Mandatory set

Presentation: group-wise > 3 team members

To mark Topic:

Regular Expression

Lecture-2

08.10.2026

(a∪b) (a∩b)

mathematical term

terms to remember

* = 0 or more

+ = 1 or more

for example:

Regular Expression: a(b)*

↓

Regular Language: starts with a followed by 0 or more b's

Definition: Regular expression is a method or representation technique which is used to ~~match~~ match pattern or strings in compiler design. Regular expression is easy to understand and have efficient implementation.

Applications:

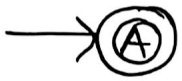
~~x~~ → Regular expression NFA use করে express করা হয়
→ যেকোন NFA diagram এর সব কিছু condition based করা যায়।

Formal Definition: Formal definition of regular expression can be expressed in 6 cases:

Case-1: R.E: $R = a,\ a \in \Sigma$



Case:2 R.E.: $R = \varepsilon$, which means string length is 0.

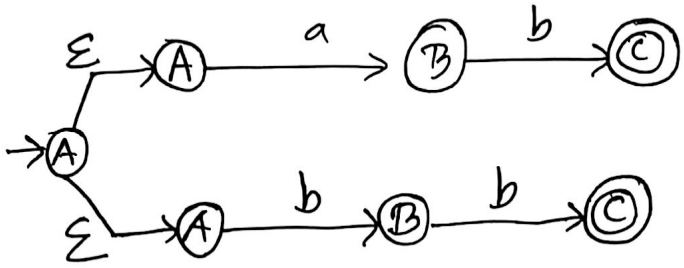


Case-3: R.E: $R = \phi$ (empty), which means string length is empty.

Case-4: (Union) $R_1 = R_1 \cup R_2$ or $R_1 + R_2$ or $R_1 | R_2$

$R_1 = ab$

$R_2 = bb$

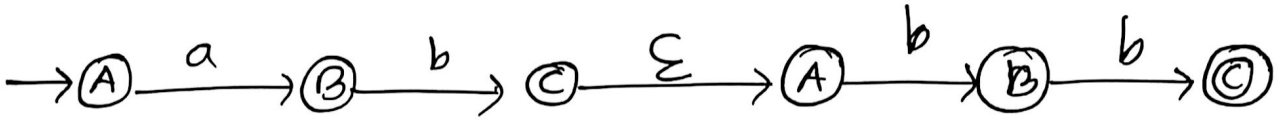


steps i) extra একটা initial state নিতে হবে। দুইটার আগে place করে connect করতে হবে।

ii) ε edge দিয়ে দুইটাকে connect করতে হবে।

Case-5 (concatenation): $R = R_1 \circ R_2$ or $R = R_1 R_2$

$R_1 = ab$ $\quad$ $R_2 = bb$



Steps: i) final state তৈরি যাবে

ii) 2nd টার initial এর সাথে connect হবে।

iii) ε দিয়ে দুই state connect হবে।

iv) যতগুলো থাকবে continue হবে।

$R = R_1 \circ R_2$, to do this R_2 remains unchanged (partial change occurs). But the final states of R_1 become non-final states and ~~R_2~~ connect to R_2 with ε edge.

Case-6 (kleene clouser)*: $R = a$

$R^* = (a)^*$

i) take a final state and initial state combo at begin with a new state name.

ii) connect the f. state to prev. initial state with ε edge.

$R = a \Rightarrow$



$R^* = a^* \Rightarrow$

